EX: 7	Information gathering using tools like Robtex
DATE:	

AIM:

To perform **information gathering and reconnaissance** on a target domain using the Robtex tool and analyze DNS records, IP information, and network relationships.

TOOLS REQUIRED

- Internet connection
- Web browser
- Robtex

PROCEDURE

1. Open a web browser.
2. Go to the Robtex website.
3. Enter the **target domain name or IP address** in the search bar.
4. Click the **search/analyze** option.
5. Robtex collects and displays information about the domain.
6. Observe the DNS records and IP information.
7. Analyze relationships such as:
    - DNS records
    - Name servers
    - Hosting information
    - Reverse DNS
8. Document the gathered information for analysis.

STEPS

1. Launch a web browser.
2. Navigate to the Robtex website.
3. Enter the target domain (example: example.com).

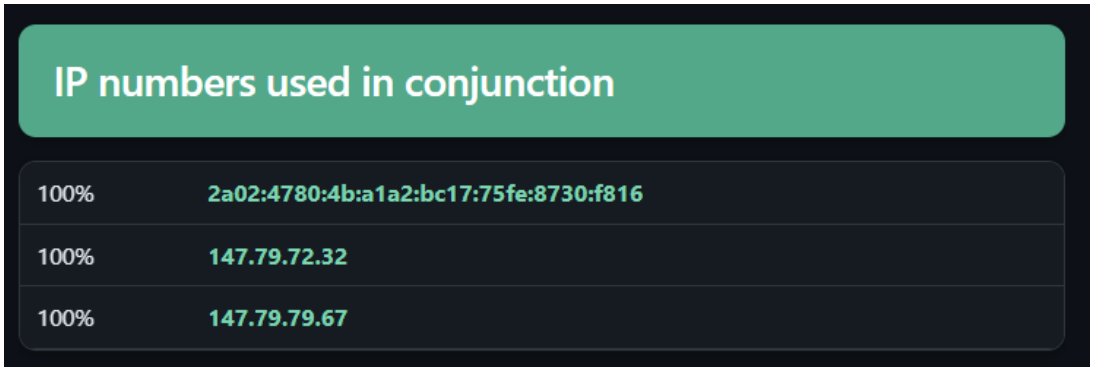
4. Click **Search**.
5. View the results including:
    - **IP address**
    - **DNS records**
    - **Name servers**
    - **Mail servers**
    - **Network routes**
6. Analyze the data displayed by Robtex.
7. Record important information for the report.

Part A – Domain: skcet.ac.in

Target Domain: skcet.ac.in

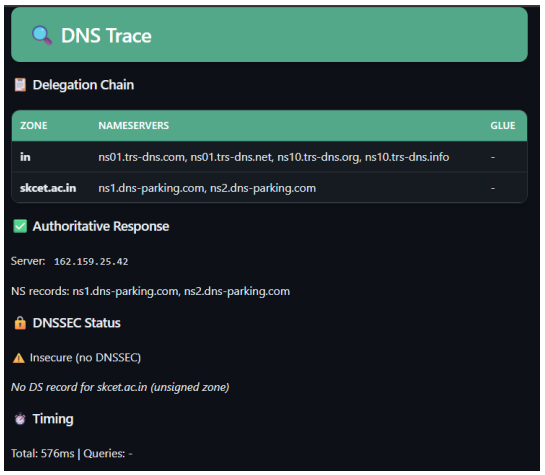
- IP Address

Robtex analysis shows the IP address associated with the domain **skcet.ac.in**. The IP address represents the server where the website is hosted.



- DNS Records

The DNS records for the domain include different record types such as **A record, NS record, and MX record** which are responsible for domain resolution and email routing.



DNS Trace

Delegation Chain

ZONE	NAMESERVERS	GLUE
in	ns01.trs-dns.com, ns01.trs-dns.net, ns10.trs-dns.org, ns10.trs-dns.info	-
skcet.ac.in	ns1.dns-parking.com, ns2.dns-parking.com	-

Authoritative Response

Server: 162.159.25.42

NS records: ns1.dns-parking.com, ns2.dns-parking.com

DNSSEC Status

Insecure (no DNSSEC)

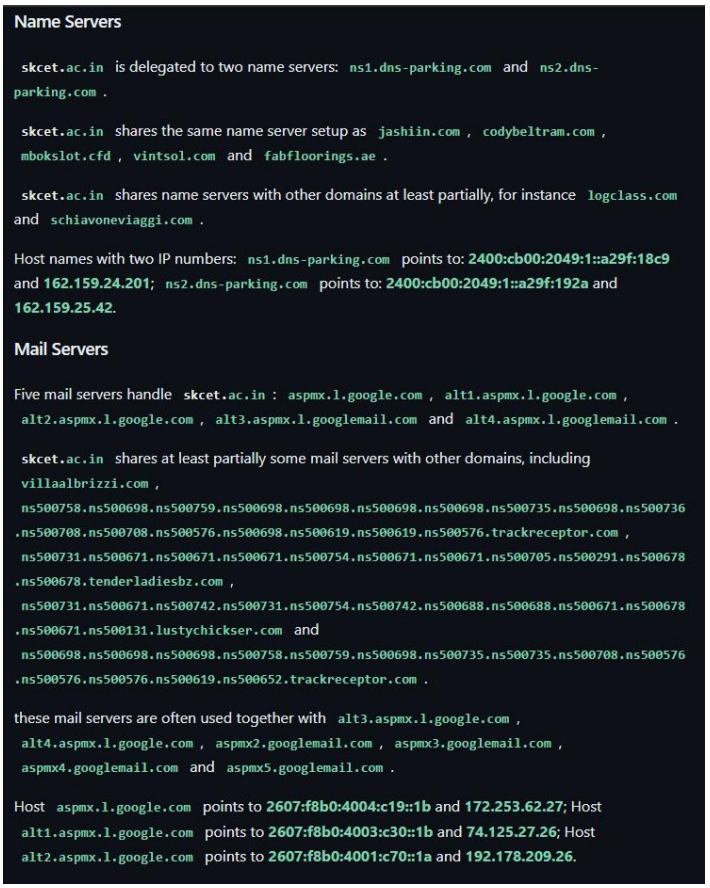
No DS record for skcet.ac.in (unsigned zone)

Timing

Total: 576ms | Queries: -

- Name Servers

The name servers manage the DNS resolution of the domain and direct traffic to the correct server.



Name Servers

skcet.ac.in is delegated to two name servers: ns1.dns-parking.com and ns2.dns-parking.com .

skcet.ac.in shares the same name server setup as jashiin.com , codybeltram.com , mbokslot.cfd , vintsol.com and fabfloorings.ae .

skcet.ac.in shares name servers with other domains at least partially, for instance logclass.com and schiavoneviaggi.com .

Host names with two IP numbers: ns1.dns-parking.com points to: 2400:cb00:2049:1::a29f:18c9 and 162.159.24.201; ns2.dns-parking.com points to: 2400:cb00:2049:1::a29f:192a and 162.159.25.42.

Mail Servers

Five mail servers handle skcet.ac.in : aspmx.l.google.com , alt1.aspmx.l.google.com , alt2.aspmx.l.google.com , alt3.aspmx.l.googlemail.com and alt4.aspmx.l.googlemail.com .

skcet.ac.in shares at least partially some mail servers with other domains, including villaalbrizzi.com , ns500758.ns500698.ns500759.ns500698.ns500698.ns500698.ns500698.ns500735.ns500698.ns500736.ns500708.ns500708.ns500576.ns500698.ns500619.ns500619.ns500576.trackreceptor.com , ns500731.ns500671.ns500671.ns500671.ns500754.ns500671.ns500671.ns500705.ns500291.ns500678.ns500678.tenderladiesbz.com , ns500731.ns500671.ns500742.ns500731.ns500754.ns500742.ns500688.ns500688.ns500671.ns500678.ns500671.ns500131.lustychickser.com and ns500698.ns500698.ns500698.ns500758.ns500759.ns500698.ns500735.ns500735.ns500708.ns500576.ns500576.ns500576.ns500619.ns500652.trackreceptor.com .

these mail servers are often used together with alt3.aspmx.l.google.com , alt4.aspmx.l.google.com , aspmx2.googlemail.com , aspmx3.googlemail.com , aspmx4.googlemail.com and aspmx5.googlemail.com .

Host aspmx.l.google.com points to 2607:f8b0:4004:c19::1b and 172.253.62.27; Host alt1.aspmx.l.google.com points to 2607:f8b0:4003:c30::1b and 74.125.27.26; Host alt2.aspmx.l.google.com points to 2607:f8b0:4001:c70::1a and 192.178.209.26.

- Mail Servers

Mail servers are responsible for handling email services associated with the domain.

Mail Servers

Five mail servers handle skcet.ac.in : aspmx.l.google.com , alt1.aspmx.l.google.com , alt2.aspmx.l.google.com , alt3.aspmx.l.googlemail.com and alt4.aspmx.l.googlemail.com .

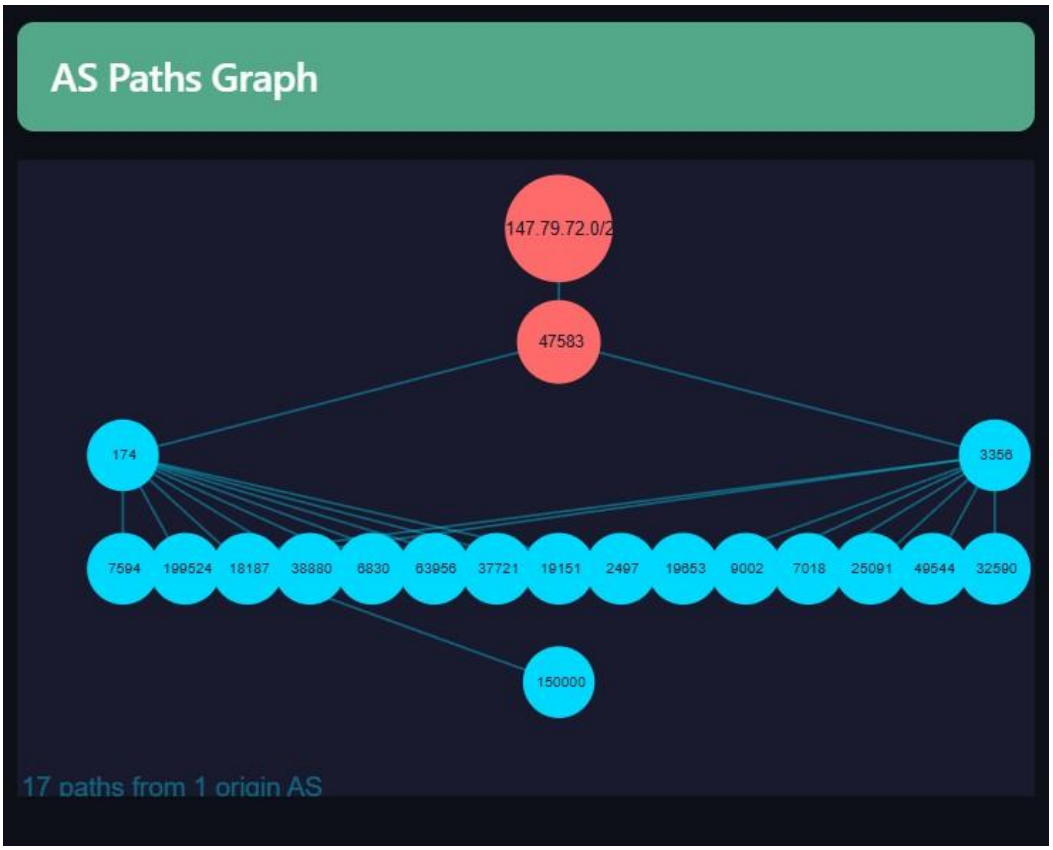
skcet.ac.in shares at least partially some mail servers with other domains, including villaalbrizzi.com , ns500758.ns500698.ns500759.ns500698.ns500698.ns500698.ns500698.ns500735.ns500698.ns500736.ns500708.ns500708.ns500576.ns500698.ns500619.ns500619.ns500576.trackreceptor.com , ns500731.ns500671.ns500671.ns500671.ns500754.ns500671.ns500671.ns500705.ns500291.ns500678.ns500678.tenderladiesbz.com , ns500731.ns500671.ns500742.ns500731.ns500754.ns500742.ns500688.ns500688.ns500671.ns500678.ns500671.ns500131.lustychickser.com and ns500698.ns500698.ns500698.ns500758.ns500759.ns500698.ns500735.ns500735.ns500708.ns500576.ns500576.ns500576.ns500619.ns500652.trackreceptor.com .

these mail servers are often used together with alt3.aspmx.l.google.com , alt4.aspmx.l.google.com , aspmx2.googlemail.com , aspmx3.googlemail.com , aspmx4.googlemail.com and aspmx5.googlemail.com .

Host aspmx.l.google.com points to **2607:f8b0:4004:c19::1b** and **172.253.62.27**; Host alt1.aspmx.l.google.com points to **2607:f8b0:4003:c30::1b** and **74.125.27.26**; Host alt2.aspmx.l.google.com points to **2607:f8b0:4001:c70::1a** and **192.178.209.26**.

- AS Paths Graph

Robtex also shows the network path and routing information between the client and the hosting server.

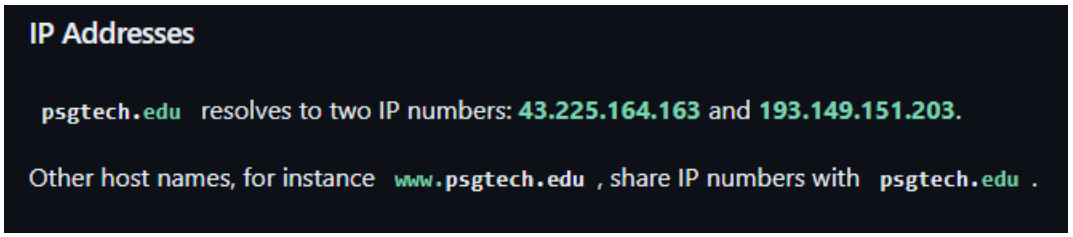


Part B – Domain: psgtech.edu

Target Domain: psgtech.edu

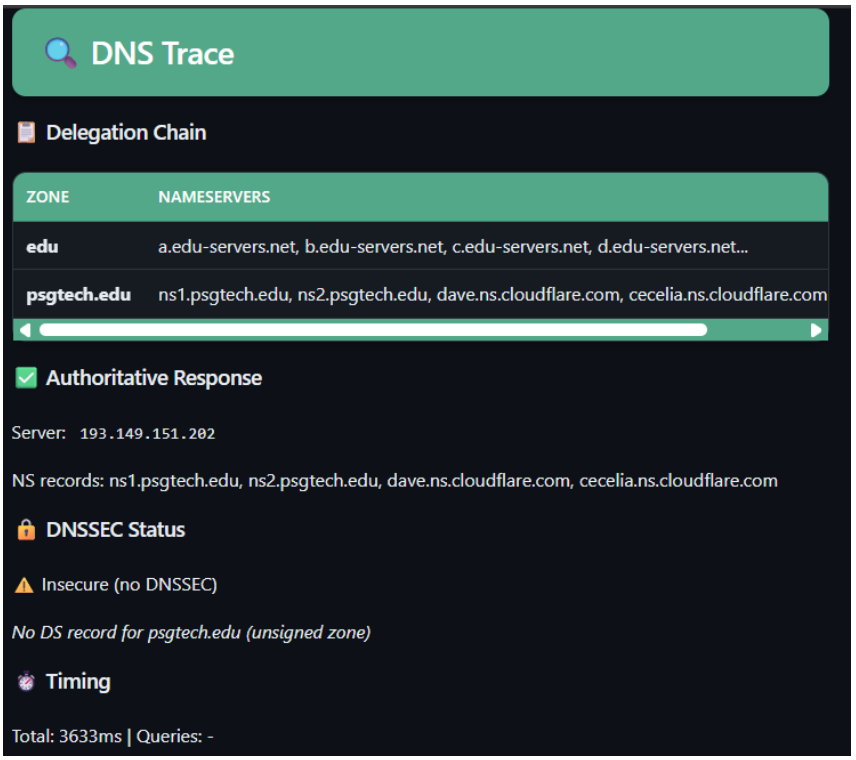
- IP Address

Robtex analysis shows the IP address associated with the domain **skcet.ac.in**. The IP address represents the server where the website is hosted.



- DNS Records

The DNS records for the domain include different record types such as **A record, NS record, and MX record** which are responsible for domain resolution and email routing.



- Name Servers

The name servers manage the DNS resolution of the domain and direct traffic to the correct server.

Name Servers

psgtech.edu is delegated to four name servers: ns1.psgtech , ns2.psgtech , ns1.psgtech.edu and ns2.psgtech.edu .

psgtech.edu at least partially shares name servers with other domains, for instance psgias.ac.in , psgsjhss.edu.in , psgps.edu.in , kssc.edu.in and psgfoundry.in .

These name servers are commonly used with the following name servers: backns1chn.vsnl.com , ns1.bsnl.in , ns3.bsnl.in , pun1lgn-a.sancharnet.in , pun2lgn-a.sancharnet.in , dns.vsnl.net.in , abts-tn-static-013.184.145.203.airtelbroadband.in , abts-tn-static-032.184.145.203.airtelbroadband.in , ns1.psgtech.ac.in and ns2.psgtech.ac.in .

Host names with one IP number:

ns1.psgtech.edu points to **193.149.151.202**

ns2.psgtech.edu points to **103.224.33.57**

- Mail Servers

Mail servers are responsible for handling email services associated with the domain.

Mail Servers

psgtech.edu is handled by five mail servers: aspmx.l.google.com , alt1.aspmx.l.google.com , alt2.aspmx.l.google.com , alt3.aspmx.l.google.com and alt4.aspmx.l.google.com .

psgtech.edu shares some mail servers with other domains, at least partially, for instance kirstywhybrow.com , ns500698.ns500758.ns500758.ns500698.ns500698.ns500736.ns500735.ns500698.ns500698.ns500708.ns500619.ns500619.trackreceptor.com , ns500759.ns500758.ns500736.ns500736.ns500736.ns500736.ns500735.ns500735.ns500708.ns500576.ns500619.ns500652.trackreceptor.com , ns500765.ns500765.ns500693.ns500734.ns500728.ns500683.ns500683.ns500683.ns500683.ns500683.ns500612.ns500110.dtrkdll.com and grabiner.com .

These mail servers are commonly used alongside aspmx2.googlemail.com and aspmx3.googlemail.com .

Host names with two IP numbers:

aspmx.l.google.com points to **2607:f8b0:4004:c19::1b** and **172.253.62.27**

alt1.aspmx.l.google.com points to **2607:f8b0:4003:c30::1b** and **74.125.27.26**

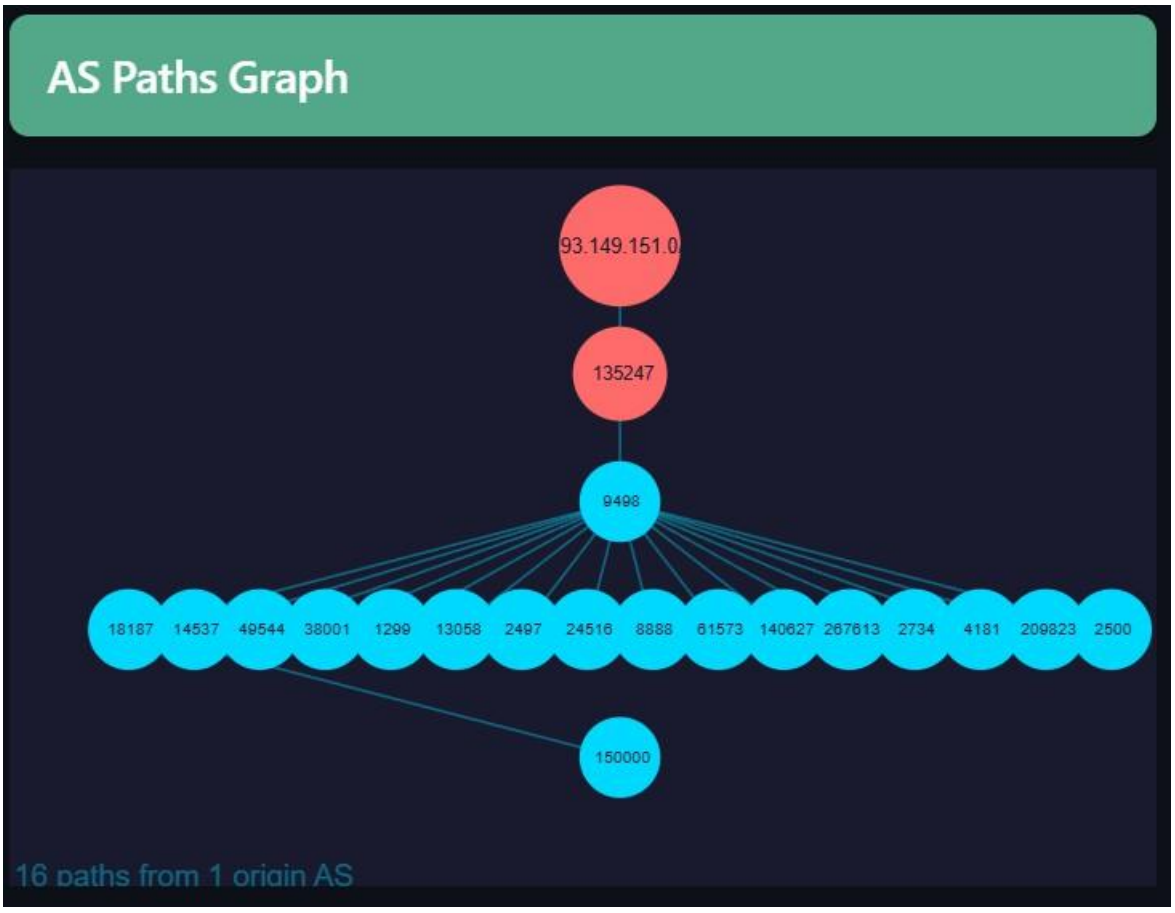
alt2.aspmx.l.google.com points to **2607:f8b0:4001:c70::1a** and **192.178.209.26**

alt3.aspmx.l.google.com points to **2607:f8b0:4024:c0d::1b** and **192.178.152.26**

alt4.aspmx.l.google.com points to **2607:f8b0:4023:403::1a** and **142.250.96.27**

- AS Paths Graph

  Robtex also shows the network path and routing information between the client and the hosting server.



Part C – Domain: kct.ac.in

Target Domain: kct.ac.in

- IP Address

  Robtex analysis shows the IP address associated with the domain **skcet.ac.in**. The IP address represents the server where the website is hosted.

- DNS Records

The DNS records for the domain include different record types such as **A record, NS record, and MX record** which are responsible for domain resolution and email routing.

DNS Trace

Delegation Chain

ZONE	NAMESERVERS	GLUE
in	ns01.trs-dns.com, ns01.trs-dns.net, ns10.trs-dns.org, ns10.trs-dns.info	-
kct.ac.in	trevor.ns.cloudflare.com, margaret.ns.cloudflare.com	-

Authoritative Response

Server: `172.64.35.154`

NS records: trevor.ns.cloudflare.com, margaret.ns.cloudflare.com

DNSSEC Status

Insecure (no DNSSEC)

No DS record for kct.ac.in (unsigned zone)

Timing

Total: 1100ms | Queries: -

- Name Servers

The name servers manage the DNS resolution of the domain and direct traffic to the correct server.

Name Servers

`kct.ac.in` is delegated to two name servers `margaret.ns.cloudflare.com` and `trevor.ns.cloudflare.com` .

`kct.ac.in` shares the same name server setup as other domains, for instance `directoryscience.com` , `oregonimmigrationresource.org` , `garmin.co.in` , `paschicchic.com` and `thiswintermachine.com` .

`kct.ac.in` at least partially shares name servers with other domains, for instance `pam2000.com` `aferp.org` , `chandigarhstockexchange.com` , `tecodiag.com` and `(0x706f726e)o-(0x736578)e.biz` .

These name servers are commonly used alongside `barbara.ns.cloudflare.com` and `armfazh.ns.cloudflare.com` .

Host names with six IP numbers:

Host name `margaret.ns.cloudflare.com` points to: **2606:4700:50::a29f:266b**, **2803:f800:50::6ca2:c26b**, **2a06:98c1:50::ac40:226b**, **108.162.194.107**, **162.159.38.107** and **172.64.34.107**.

Host name `trevor.ns.cloudflare.com` points to: **2606:4700:58::a29f:2c9a**, **2803:f800:50::6ca2:c39a**, **2a06:98c1:50::ac40:239a**, **108.162.195.154**, **162.159.44.154** and **172.64.35.154**.

- Mail Servers

Mail servers are responsible for handling email services associated with the domain.

Mail Servers

kct.ac.in is handled by five mail servers: d233344a.ess.barracudanetworks.com , d233344b.ess.barracudanetworks.com , kctbacu.kct.ac.in , kct-ac-in.mail.protection.outlook.com and _dc-mx.459cddebc62a.kct.ac.in .

Eight IPs point to kct-ac-in.mail.protection.outlook.com : **2a01:111:f403:cc2c::**, **2a01:111:f403:cc2c::1**, **2a01:111:f403:cc2d::**, **2a01:111:f403:cc2d::1**, **52.101.144.0**, **52.101.144.3**, **52.101.145.0** and **52.101.145.2**.

Three IPs point to d233344a.ess.barracudanetworks.com : **209.222.82.252**, **209.222.82.253** and **209.222.82.255**. The same set also points to d233344b.ess.barracudanetworks.com : **209.222.82.252**, **209.222.82.253** and **209.222.82.255**.

One IP points to kctbacu.kct.ac.in : **103.196.28.200**. One IP points to _dc-mx.459cddebc62a.kct.ac.in : **203.193.157.140**.

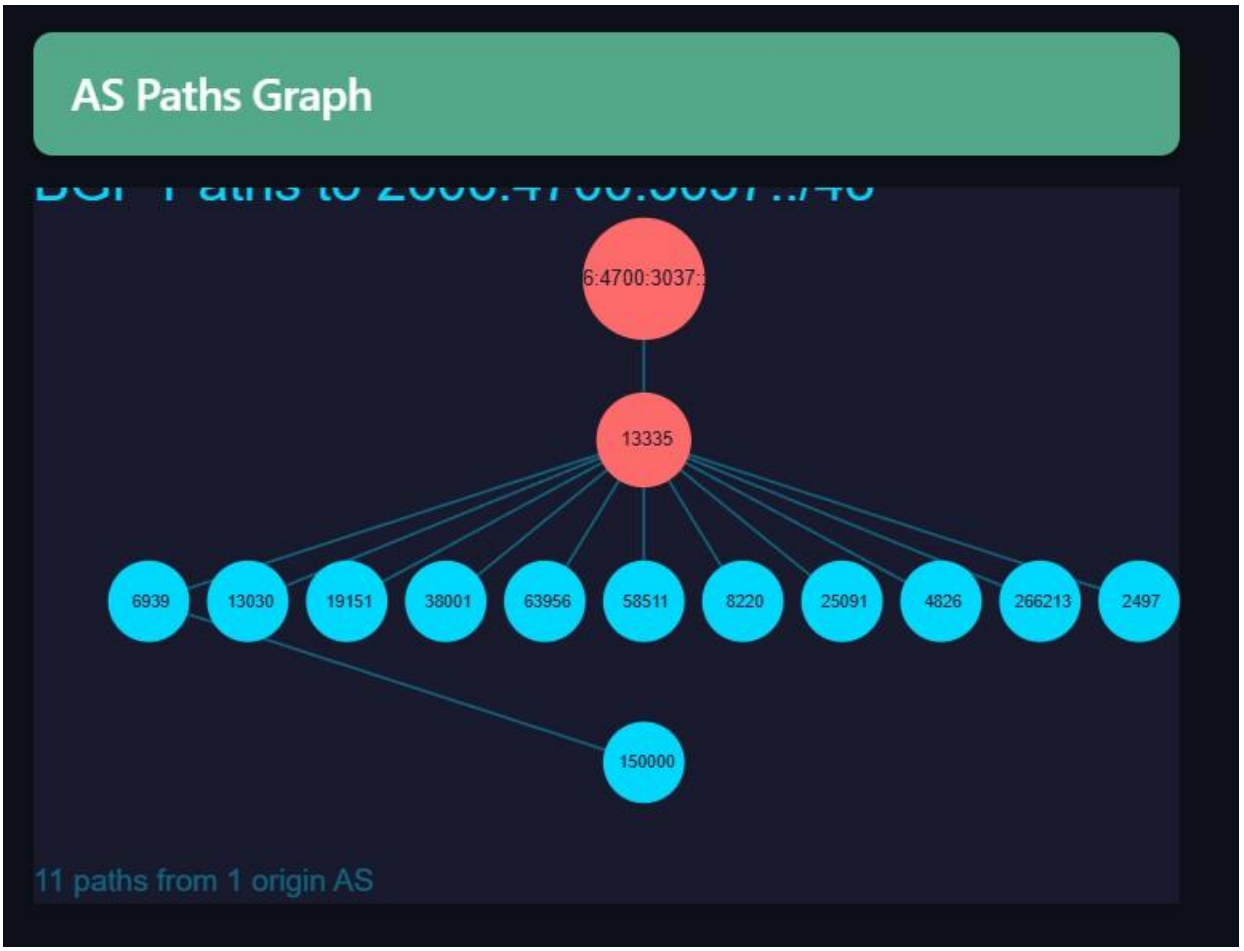
Hosts that point to **209.222.82.252**: d233344a.ess.barracudanetworks.com and d233344b.ess.barracudanetworks.com .

Hosts that point to **209.222.82.253**: d233344a.ess.barracudanetworks.com and d233344b.ess.barracudanetworks.com .

Hosts that point to **209.222.82.255**: d233344a.ess.barracudanetworks.com and d233344b.ess.barracudanetworks.com .

- AS Paths Graph

  Robtex also shows the network path and routing information between the client and the hosting server.



Result:

Information gathering and reconnaissance were successfully performed on the target domains **skcet.ac.in, psgtech.edu, and kct.ac.in** using the **Robtex** tool. The analysis provided detailed insights into the domain infrastructure including **IP address mapping, DNS records, name servers, mail servers, and network routing information**.